1. Error and inverse stability

1. Pointwise bias-variance decomposition

Fix x_0 and let the random training sample S produce $\hat{f}_S(x_0)$. A fresh response is

$$Y = f^\star(x_0) + \varepsilon, \quad f^\star(x_0) = \mathbb{E}[Y|x_0].$$

Assume squared loss, $\mathbb{E}[\varepsilon|x_0] = 0$, $\mathrm{Var}(\varepsilon|x_0) = \sigma^2(x_0)$, and fresh ε independent of S. With $\mu(x_0) = \mathbb{E}_S\left[\hat{f}_S(x_0)\right]$,

$$\mathbb{E}_{S,\varepsilon}\left[\left(Y - \hat{f}_S(x_0)\right)^2\right] = (\mu - f^\star)^2 + \mathrm{Var}_S\left(\hat{f}_S(x_0)\right) + \sigma^2(x_0).$$

The three terms are squared systematic displacement, refit spread, and fresh noise. The identity is **pointwise**: x_0 is fixed.

2. Why the cross terms vanish

At the fixed point, write $Y - \hat{f} = (f^\star - \mu) + \left(\mu - \hat{f}\right) + \varepsilon$.
The training-sample cross term is zero because $\mathbb{E}_S\left[\hat{f}\right] = \mu$. For a fresh-noise cross term, condition first on the sample:

$$\mathbb{E}\left[\left(\mu - \hat{f}\right)\varepsilon\right] = \mathbb{E}_S\left[\left(\mu - \hat{f}\right)\mathbb{E}[\varepsilon|S, x_0]\right] = 0.$$

Freshness plus zero conditional mean supplies $\mathbb{E}[\varepsilon|S, x_0] = 0$. $\mathbb{E}[\varepsilon] = 0$ alone does **not** rule out correlation with the fitted predictor; if freshness fails, extra terms may remain.

3. Hadamard audit: three separate tests

An inverse problem is well posed on its declared admissible spaces only if:

1. a solution exists for every admissible observation;
2. the solution is unique in the modeled solution class;
3. the solution depends continuously on the observation.

Failure of any one makes the problem ill posed. For $\mathrm{diag}(1, 0)$, uniqueness or existence can fail. For $\mathrm{diag}(1, 0.01)$, the inverse exists and is continuous, but has norm 100: severe sensitivity, not discontinuity. **Zero** and **small positive** singular values are qualitatively different.

4. Singular-value conditioning

For singular pairs $Av_j = \sigma_j u_j$, $\sigma_j \geq 0$, every active direction $\sigma_j > 0$ is inverted by

$$\langle f, v_j\rangle = \frac{\langle y, u_j\rangle}{\sigma_j}.$$

An observation perturbation in direction u_j is amplified by $\frac{1}{\sigma_j}$. For invertible A,

$$\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}.$$

κ_2 is a worst-case **relative** sensitivity measure; $\frac{1}{\sigma_{\min}}$ is an **absolute** inverse-amplification bound. A condition number is not the statistical variance from the bias-variance decomposition, though small singular directions provide a mechanism for refit instability.

2. Tikhonov regularization

1. Objective and finite-dimensional solution

For $A : \mathcal{H} \to \mathcal{K}$, observation y, and $\lambda > 0$, zero-prior Tikhonov solves

$$f_\lambda \in \arg\min_f \{\|Af - y\|^2 + \lambda\|f\|^2\}.$$

The first term is data misfit; the second charges for solution size. Increasing λ changes their relative weight; it does not remove noise from y.
For $A \in \mathbb{R}^{n\times d}$ with Euclidean norms,

$$(A^TA + \lambda I)f_\lambda = A^Ty.$$

For nonzero z, $z^T(A^TA + \lambda I)z = \|Az\|^2 + \lambda\|z\|^2 > 0$. Thus the finite-dimensional linear quadratic has exactly one minimizer even when A has a null space. This proof does not establish a general nonlinear or infinite-dimensional existence theorem.

2. Directionwise filter and stability

Using the positive compact-SVD pairs $Av_j = \sigma_j u_j$,

$$f_\lambda = \sum_{j=1}^r \frac{\sigma_j}{\sigma_j^2 + \lambda}\langle y, u_j\rangle v_j, \quad P_{\ker(A)}f_\lambda = 0.$$

- unregularized data-to-solution gain: $\frac{1}{\sigma_j}$;
- regularized gain: $\frac{\sigma_j}{\sigma_j^2+\lambda}$;
- fraction of the unregularized coefficient retained: $\frac{\sigma_j^2}{\sigma_j^2+\lambda}$.

The zero-prior penalty sets the separate null-space component to zero; it does not recover missing information. For fixed $\lambda > 0$,

$$\|f_\lambda(y + \Delta y) - f_\lambda(y)\|_2 \leq \frac{1}{2\sqrt{\lambda}}\|\Delta y\|_2,$$

because $\max_{\sigma\geq 0} \frac{\sigma}{\sigma^2+\lambda} = \frac{1}{2\sqrt{\lambda}}$.

3. Zero-penalty boundary

In finite-dimensional linear least squares with the Euclidean zero-prior penalty,

$$f_\lambda \to A^\dagger y \quad \text{as} \quad \lambda \to 0^+.$$

Positive singular coordinates approach the inverse coefficients, while every null-space coordinate remains zero. The limit is therefore the Moore-Penrose minimum-norm least-squares solution: the minimum-norm exact solution if the system is consistent, or the minimum-norm member among least-squares minimizers otherwise. Under full column rank this reduces to the unique OLS solution.
For general inverse problems, sending λ to zero is not enough. A representative noise-compatible schedule also requires $\frac{\delta^2}{\lambda} \to 0$, together with the relevant analytic assumptions. Fixed positive λ is not an exact-recovery limit.

3. Ridge specialization

1. Convention, objective, estimator

Let $X \in \mathbb{R}^{n\times d}$ be fixed, with centered columns and centered response, so no intercept is displayed or penalized. Ridge minimizes

$$J_\lambda(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|^2, \quad \lambda > 0,$$

and has the registered estimator

$$\hat\beta_\lambda = (X^T X + \lambda I_d)^{-1} X^T y.$$

The Hessian is $2(X^T X + \lambda I_d)$ and is positive definite, so the coefficient minimizer is unique even for dependent columns. Uniqueness alone does not certify lower prediction risk.

Scaling: if a source uses $\frac{\|y-X\beta\|^2}{n} + \lambda'\|\beta\|^2$, the same family uses $\lambda = n\lambda'$. Do not mix their numerical parameter values. An unpenalized intercept may equivalently be fitted before centering; penalizing it is a different modeling convention.

2. Rank and selection boundaries

With compact SVD $X = U_r \Sigma_r V_r^T$,

$$\hat\beta_\lambda = \sum_{j=1}^{r} \frac{\sigma_j}{\sigma_j^2 + \lambda} \langle u_j, y\rangle v_j.$$

Hence $\hat\beta_\lambda \to X^+ y$ as $\lambda \to 0^+$. Under full column rank, this is $(X^T X)^{-1} X^T y$; under rank deficiency it is the minimum-norm OLS minimizer, not an arbitrary OLS solution. As $\lambda \to \infty$, every displayed coefficient tends to zero.

Ridge continuously shrinks generic coefficients and does not search feature subsets or normally set coefficients exactly to zero. A coefficient in an exact design-null direction is zero because it cannot improve fit but still pays the penalty—not because ridge performed variable selection.

4. Directional bias and variance

1. Gram eigendirections and shrinkage

Let

$$G = X^T X = Q\,\mathrm{diag}(\mu_1, \ldots, \mu_d) Q^T, \quad \mu_j \ge 0,$$

where μ_j are squared singular values and $z_j = \langle q_j, X^T y\rangle$. Ridge uses coefficient $\frac{z_j}{\mu_j+\lambda}$. For $\mu_j > 0$, relative to OLS it retains

$$s_j(\lambda) = \frac{\mu_j}{\mu_j + \lambda}.$$

Small-μ_j directions shrink most. Keep three quantities distinct:

- $\frac{1}{\mu_j+\lambda}$ acts on z_j;
- $s_j(\lambda)$ is the fraction of the OLS coefficient retained;
- if $\mu_j = 0$, ridge assigns zero in that coefficient direction.

2. Fixed-design probability model

To discuss moments, declare

$$y = X\beta^\star + \varepsilon, \quad \mathbb{E}[\varepsilon|X] = 0, \quad \mathrm{Cov}(\varepsilon|X) = \sigma^2 I_n,$$

with fixed X and a correctly specified linear conditional mean. Then

$$\mathbb{E}\left[\hat\beta_\lambda|X\right] = (G + \lambda I)^{-1} G\beta^\star,$$

$$\mathrm{Cov}\left(\hat\beta_\lambda|X\right) = \sigma^2 (G+\lambda I)^{-1} G (G+\lambda I)^{-1}.$$

For $\theta_j = \langle q_j, \beta^\star\rangle$,

$$\mathbb{E}\left[\hat\theta_{\lambda,j}|X\right] = s_j(\lambda)\theta_j,$$

$$\mathrm{bias}_j = -\frac{\lambda}{\mu_j + \lambda}\theta_j,$$

$$\mathrm{variance}_j = \sigma^2 \frac{\mu_j}{\left(\mu_j + \lambda\right)^2}.$$

3. Coefficients are not predictions

Coefficient and prediction risks differ. For fitted values $X\hat\beta_\lambda$, directional squared-bias and variance gain μ_j; sum and divide by n for in-design mean error, while fresh-response error adds noise. Training residuals determine none; λ trades bias for sensitivity, so risk need not be monotone.

5. Diagnose before regularizing

1. Evidence-to-intervention map

Equal training error can hide different failures. Combine variation across refits, Gram-spectrum conditioning, systematic prediction bias, fresh-noise scale, and the scientific prediction goal.

Dominant evidence	First response
high sample variance, no weak-direction evidence	more data or another justified variance control; ridge is not automatic
variance aligned with tiny μ_j	ridge is targeted; monitor its added bias
high systematic bias	enrich the representation/model, not stronger shrinkage
high fresh-noise floor	improve measurement or accept the irreducible floor

In every case, later independent evidence should estimate out-of-sample predictive risk and be interpreted against the diagnosed failure/noise floor.

2. Diagnosis, selection, and final testing

Diagnosis says what λ must trade off; it does not choose its value. Fit candidates on training data, select using honest out-of-sample evidence or another justified criterion, then use the untouched final test set once. Tuning on final-test error makes the reported minimum optimistically biased.

The penalty is an SRM-style fit-complexity compromise across $\mathcal{H}_B = \{x \mapsto \beta^T x : \|\beta\| \le B\}$. It does not literally lower the VC dimension of an unchanged class: a penalty changes the criterion; a class restriction changes the hypothesis set.